

Cover Letter ■■ TMLR Submission (with arXiv as fallback)

To: TMLR Action Editor From: AGI Research Kit Contributors Date: 2026-08-01 Re: Single-paper submission (replaces a prior 5-paper bundle that has been archived)

Dear Editor, We are pleased to submit our manuscript, "AGI Kit: An End-to-End Self-Improving Tool-Use Pipeline on Consumer Hardware ■■ Empirical Observations", for consideration at TMLR. The submission is a single unified preprint; a prior 5-paper bundle has been archived at `papers/_deprecated/` and may be inspected for audit but is no longer the submission of record. The paper reports what we built and what we measured: a four-layer self-improving tool-use pipeline (per-step reflection, semantic strategy memory, continual learning with an A/B safety gate, and bounded recursive self-modification) running entirely on consumer hardware (CPU-only, ~5 GB RAM) with a Qwen3-1.7B base model. Headline numbers: End-to-end JSON-final emission: 100% on full_run3 with all four layers; correctness on the arithmetic subset: 77.6% (Sections 4.1.1-4.1.2; the 30% static baseline is the JSON-final emission rate of the bare Qwen3-1.7B on the same 50-task synthetic set) Continual-learning eval: 60.4% ■■ 3.6% across 3 seeds (t-test vs static, $t=14.6$, $p<0.01$) A/B safety gate: 12/12 adversarial boundary tests passed L4 prompt-injection red team: 18/18 malicious blocked, 0 false positives on 30 attack vectors Gate calibration across 5 deployment profiles: 60/60 trials with monotonic acceptance-rate response